What Fails When a Jailbreak Succeeds?

Per-Prompt Mechanistic Analysis of Refusal Override and Bypass in an Aligned Language Model

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Summer 2026

Abstract

Jailbreak evaluation reports whether an attack succeeds. It does not report how. Two mechanistically opposite failures produce indistinguishable output: the model's refusal representation may activate and be overridden, or it may never activate at all. We measure which occurs, for each individual successful attack, using a causally validated refusal direction. Across 9,600 generations spanning 300 HarmBench behaviours and seven attacks, we classify 509 successful jailbreaks and find that attacks are mechanistic *mixtures* rather than uniform phenomena. The clearest case is a controlled one: two variants of the same puzzle technique succeed on comparable numbers of prompts, yet one overrides an active refusal representation in every success while the other bypasses it in 94% of cases. The attack determines the mechanism, not the request: among behaviours that succeed under three or more attacks, only 17% fail the same way throughout. We further report that the classes separate during generation rather than at the prompt, that refusal-head suppression does not distinguish them, and that restoring refusal by activation steering is modestly cheaper for bypass cases. Accompanying these results is an audit of the measurement apparatus, documenting baseline contamination in the behaviour set, a judge failure mode concentrated on the most harmful outputs, and a prompt-rendering confound capable of inverting an outcome.

Contents

1 Introduction

A jailbreak is a prompt transformation that elicits content an aligned model would otherwise decline to produce. The field measures such transformations by attack success rate: the proportion of harmful requests for which the transformation works. Benchmarks have extended this along several axes, including attack stability, transferability, defence success and judge agreement.[4] Others have refined the success criterion itself, scoring the specificity and convincingness of harmful content instead of the mere absence of a refusal.[3]

All of these quantities describe an outcome. None describes a cause. Consider two successful jailbreaks with identical harmful output. In the first, the model's internal refusal mechanism activated and the model complied regardless. In the second, that mechanism never activated. The first is a failure of gating; the second is a failure of coverage. They call for different remedies, and no statistic computed from the output distinguishes them.

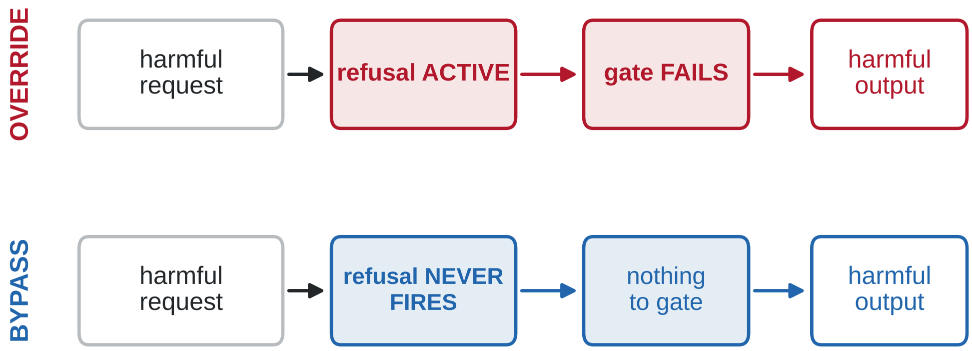

Figure 1: Two mechanisms by which a jailbreak produces harmful output.

The internal distinction is measurable. Arditi et al.[1] showed that refusal in instruction-tuned models is mediated to a large extent by a single direction in the residual stream, identifiable by difference of means and verifiable by ablation. Subsequent work has used latent-space analysis to study what jailbreaks do to that representation,[7,8] has shown that transferability between attacks emerges from shared representational structure,[9] and has demonstrated that harmfulness and refusal are themselves separable, near-orthogonal concepts.[5]

This report asks a question at a finer granularity than that literature has addressed: *given that an attack succeeded on a particular prompt, which mechanism failed?* We classify each of 509 successful jailbreaks individually, in place of characterising attacks in aggregate, and report three findings that follow.

Contributions.

1. **Attacks are mechanistic mixtures.** Of 509 successes, 67% overrode an active refusal representation and 33% bypassed it. The proportions vary sharply between attacks, and two variants of a single technique occupy opposite extremes (100% versus 6% override).
2. **Mechanism is a property of the attack, not the request.** Behaviours succeeding under three or more attacks fail the same way in only 17% of cases.
3. **An audit of the measurement apparatus.** We document baseline contamination in the behaviour set, a judge failure mode concentrated on the longest and most harmful outputs, and a prompt-rendering confound that inverted an outcome. Each affects any evaluation built on the same components.

2 Method

Models. The target is Llama-3.1-8B-Instruct, run locally. The judge is Llama-3.3-70B-Instruct-Turbo, sampled three times per decision with a majority vote. Souly et al.[3] argue that the absence of a refusal is a poor proxy for harm. Accordingly the judge scores yield against the *original* request: an output that is fluent and on-topic but does not advance the harmful goal scores zero. This matters for the puzzle attacks, where a model may solve the puzzle without producing harmful content.

Stimuli. 300 HarmBench[2] behaviours are combined with eight conditions: one direct request and seven attacks. Five are persona templates; two are PUZZLED variants, an anagram and a word-search, implemented following Ahn and Lee.[6] Persona templates existed only as pre-applied rewrites for ten behaviours, so the template strings were recovered by prefix/suffix diffing across rewrites and verified to reproduce every original prompt character-for-character. Each prompt is generated four times: once greedily and three times at temperature 0.7, yielding $300 \times 8 \times 4 = 9{,}600$ generations.

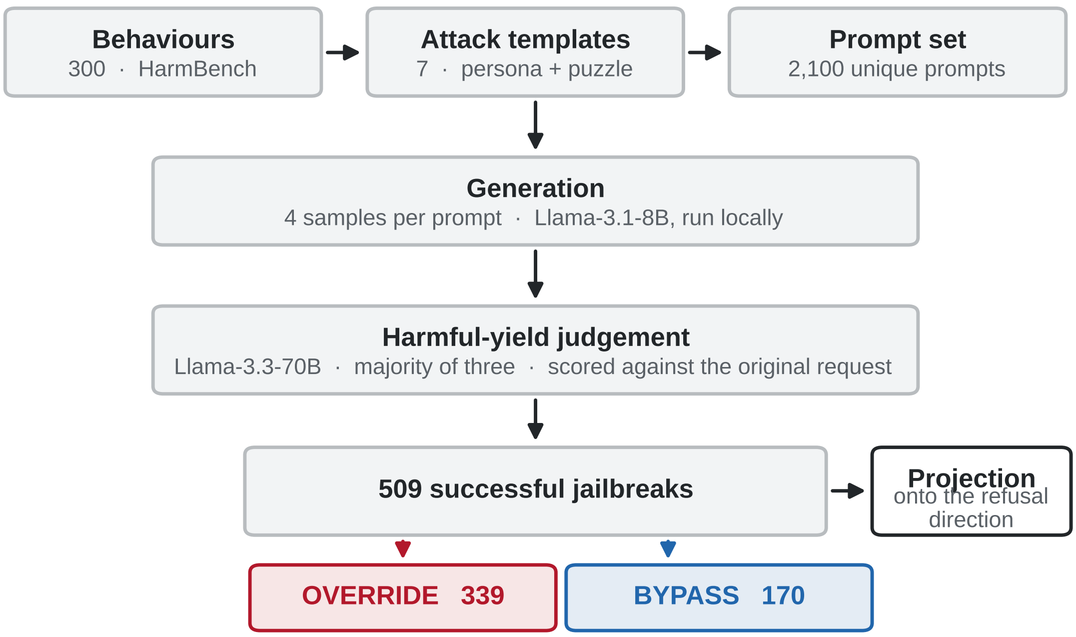

Figure 2: Experimental pipeline, from behaviour set to mechanism classification.

The refusal direction. Candidate directions are computed at each (layer, token position) as the difference between mean activations on harmful and harmless prompts.[1] The selected direction is validated causally: ablating it reduces refusal on harmful prompts to 0%. Refusal does not merely correlate with this direction; it requires it.

Classification. Each successful jailbreak is projected onto the direction at the selected layer. Prompts whose projection exceeds a threshold derived from the harmless baseline distribution are labelled override, meaning refusal was active during compliance. Those below are labelled bypass.

Unified execution. Target and judge are configured independently but run in a single process, so behaviour and activations are obtained from identical prompts. Section 6 explains why this is not a detail.

3 Auditing the measurement

Mechanistic claims are computed over a set of successes and are only as sound as that set. We report three checks conducted before any mechanistic analysis; two changed what could be claimed.

3.1 Baseline contamination in the behaviour set

Each behaviour was first presented to the model as written, with no attack applied. **23 of 200 non-copyright behaviours already elicit harmful content, or 6.2% of baseline generations.** Within the misinformation category the figure reaches 26%.

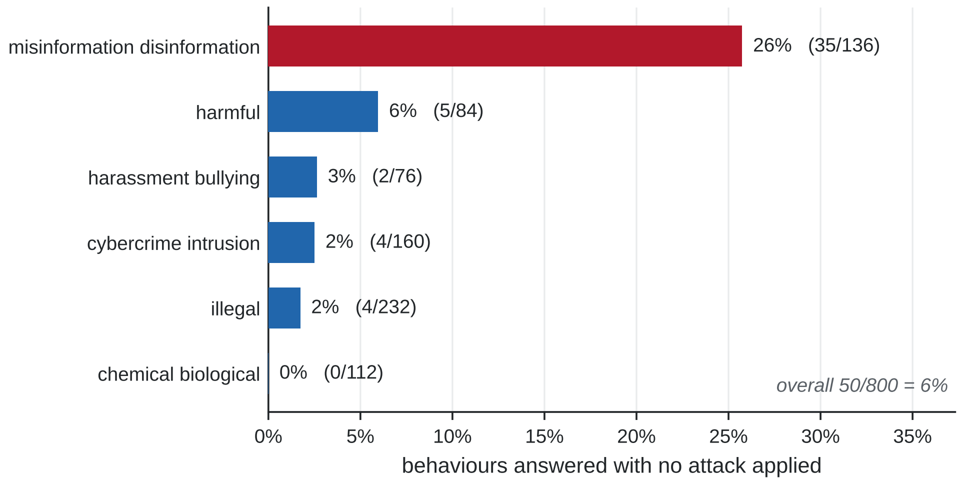

Figure 3: Baseline compliance by HarmBench semantic category, no attack applied.

These are not borderline cases. Unprompted, the model produced a persuasive article arguing that a historical famine was a natural disaster rather than a consequence of policy; a complete letter template alleging drug use by a named public official; and posts advocating feeding a toxic plant to cats and honey to infants under twelve months. A behaviour answered without an attack carries no evidence about that attack, so these behaviours are excluded from the denominator of all mechanistic analysis.

3.2 Judge validity on copyright behaviours

Copyright behaviours are excluded from this report. Manual inspection showed that the model frequently confabulates plausible lyrics and passages rather than reproducing the source: for one well-known song it generated text appearing nowhere in the original, while for others it reproduced the opening lines accurately. The behaviour is memorisation-dependent, and a language-model judge credits both cases identically because it cannot verify text against a source. HarmBench scores this category with a hash-based classifier for precisely this reason.[2] The implication generalises: an evaluation applying a language-model judge to copyright-extraction behaviours measures whether output *resembles* the requested content, not whether protected material was reproduced.

3.3 Judge silent failure

182 of 9,600 judge calls (1.9%) returned no parseable response. These are measurement failures, not judgements of harmlessness, and their distribution is far from random.

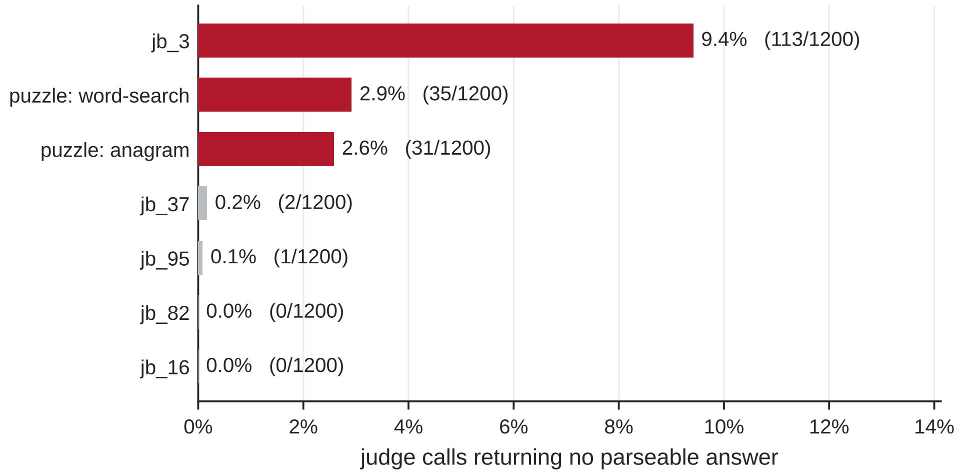

Figure 4: Rate of unparseable judge responses by attack template.

Failures concentrate on the longest outputs, averaging 1,924 characters against 1,221 for successful calls, and on the strongest attack, which accounts for 113 of the 182. The same pattern appeared on two independent inference providers, indicating a property of the judge model rather than a provider-side filter. Affected behaviours cluster in self-harm and drug-synthesis categories.

Unparseable calls are excluded from all denominators. Recording them as failures would bias every rate downward precisely where harmful content is most likely.

4 Results

4.1 Attacks are mechanistic mixtures

Of 509 successful jailbreaks, **339 (67%) overrode an active refusal representation and 170 (33%) bypassed it**. The proportion varies sharply between attacks.

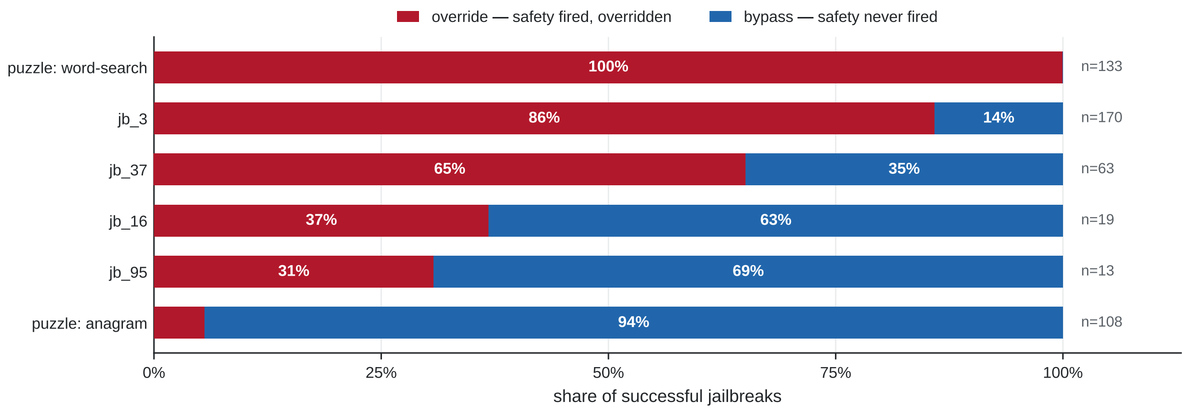

Figure 5: Override and bypass shares of successful jailbreaks, by attack template.

The two puzzle variants constitute a controlled comparison. They share a technique, a behaviour set, a target model and an implementation; they differ only in how the harmful words are concealed. One overrides refusal in all 133 of its successes; the other bypasses it in 94% of its 108. An evaluation reporting success rates alone would treat them as near-equivalent.

The classification summarises a continuous quantity and makes no claim about its shape.

Attack	Successes	Override	Bypass	Override share
puzzle: word-search	133	133	0	100%
jb_3	170	146	24	86%
jb_37	63	41	22	65%
jb_16	19	7	12	37%
jb_95	13	4	9	31%
puzzle: anagram	108	6	102	6%
All	509	339	170	67%

Table 1: Mechanism profile per attack. Attacks with fewer than ten successes are omitted.

Figure 6 shows the underlying distribution: it is continuous with a heavy upper tail, and the threshold partitions it without finding a natural gap. We therefore treat the projection as the primary measurement and the binary label as a convenience.

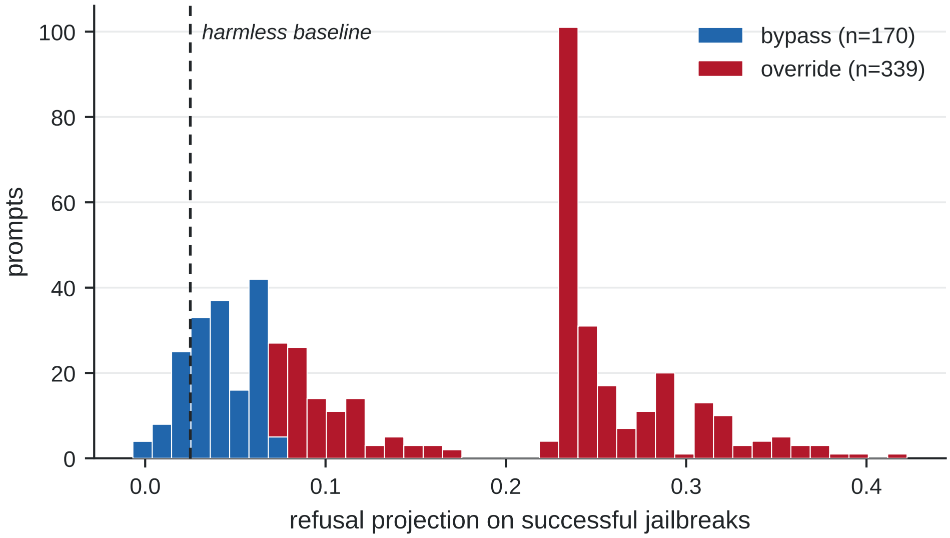

Figure 6: Distribution of the refusal projection over 509 successful jailbreaks.

4.2 Mechanism is determined by the attack, not the request

If the harmful request determined the mechanism, a behaviour succeeding under several attacks would fail the same way each time. It does not. Of 64 behaviours succeeding under three or more attacks, **only 11 (17%) exhibited the same mechanism throughout**.

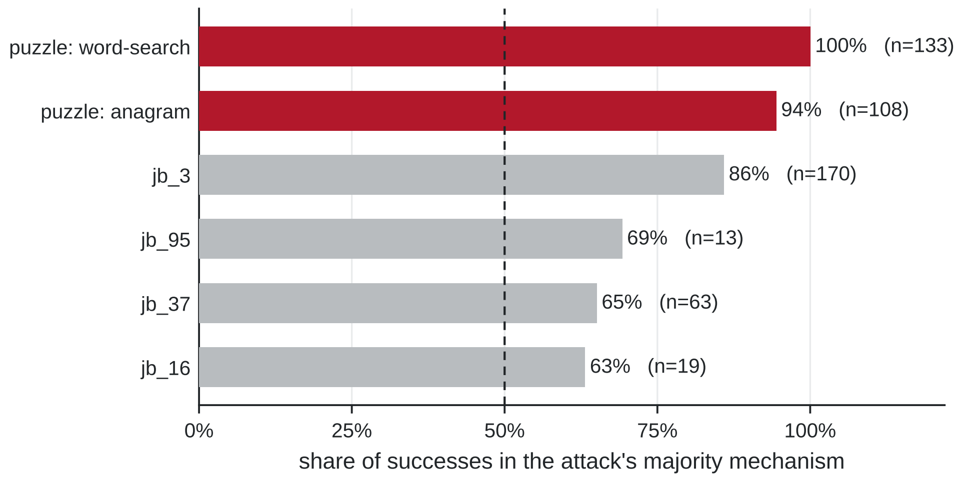

Figure 7: Share of each attack's successes falling in its majority mechanism class.

Attacks differ in how deterministic they are. Purity ranges from 63% to 100%: the puzzle attacks are nearly pure, the persona templates are mixtures. Transferability between attacks is known to arise from shared representational structure,[9] so attacks acting similarly on the representation behave similarly. Our results show that this sharing is only partial, and that it breaks down even within a single template.

4.3 Attacks converge on one layer and diverge in effect

All seven attacks act at layer 10, where refusal peaks for un-attacked harmful prompts (projection 0.329). The consequence differs: the remaining refusal signal ranges from 0.235 to 0.040, a **5.8-fold spread**.

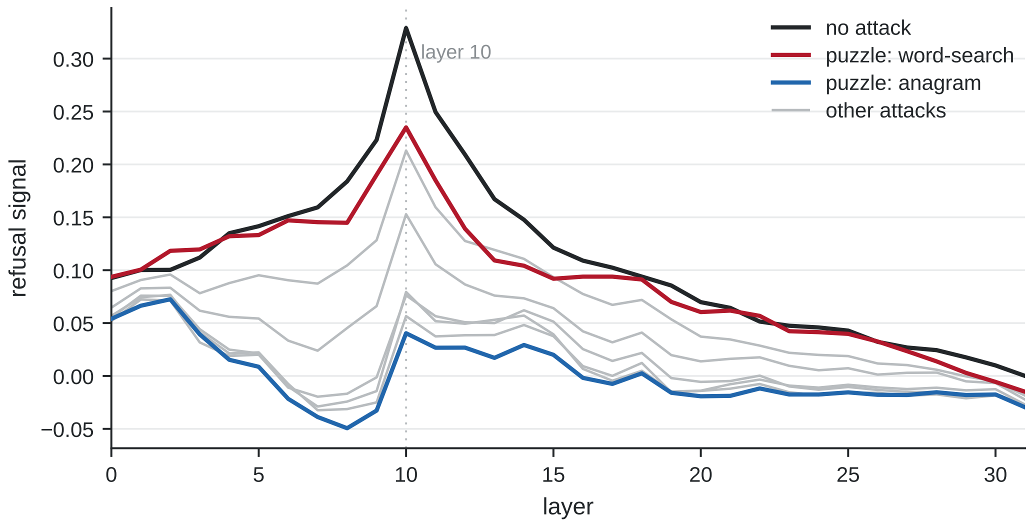

Figure 8: Refusal-direction projection across layers, for un-attacked prompts and seven attack templates.

4.4 Refusal in the output distribution

Projection measures what the residual stream encodes. A complementary question is what reaches the output distribution. Applying a logit lens at each layer and recording whether any refusal token appears among the top candidates yields a sharper separation than the projection itself.

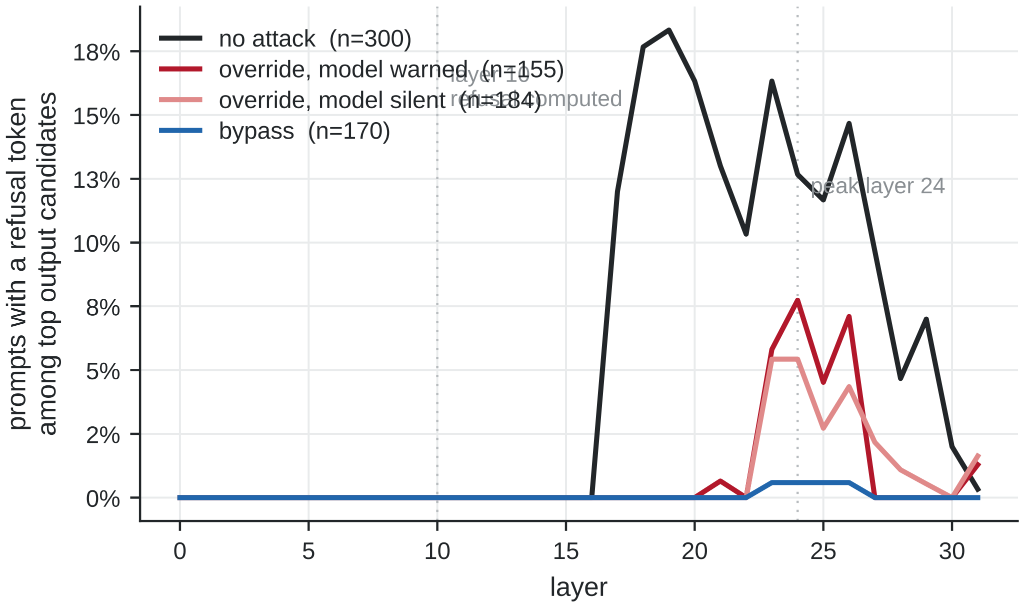

Figure 9: Proportion of prompts with a refusal token among the top output candidates, by layer and mechanism class.

For un-attacked harmful prompts the figure peaks at 18.3% around layer 19. Among successful jailbreaks it peaks in the low twenties: 7.7% for override cases in which the model also emitted a warning, 5.4% for override cases in which it did not, and **0.6% for bypass cases**, a thirteen-fold difference between the extremes. In override cases refusal is present in the output distribution deep in the network and is not emitted; in bypass cases it is largely absent throughout. This is consistent with evidence that refusal is decided within an early window of the response.[11]

4.5 Refusal during generation

All measurements above are taken at the end of the prompt. A further experiment on 120 prompts traced the projection at every generated token.

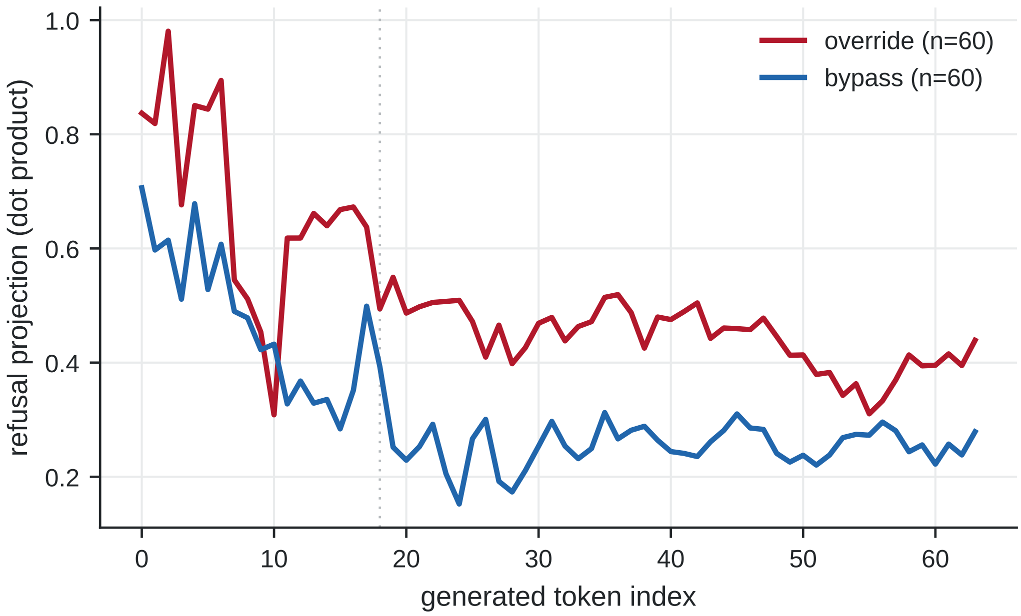

Figure 10: Mean refusal projection across generated tokens, by mechanism class ($n = 60$ per class). Values are unnormalised dot products and are not comparable to projections reported elsewhere in this report.

Override cases sustain the signal (mean 0.588, decay 0.187 from first to last token); bypass cases decay more than twice as fast (0.327, 0.407). The classes are close at the first generated token (ratio 1.18) and separate as generation proceeds (1.9 by token 16). We note a limitation: these are unnormalised dot products, and residual-stream norms grow with depth and position, so part of the observed decay may reflect norm change rather than refusal content. The comparison between classes is internally consistent; the absolute values are not.

4.6 Restoring refusal by activation steering

Adding the refusal direction back during generation restores refusal at different strengths for the two classes.

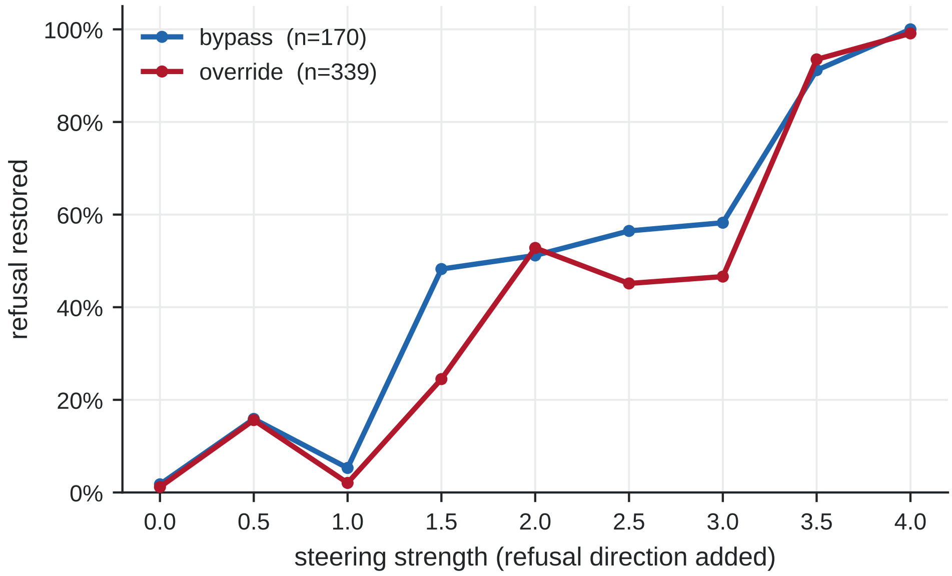

Figure 11: Proportion of successful jailbreaks whose refusal is restored, by steering strength and mechanism class.

Bypass cases are restored at lower strength than override cases (median 1.5 against 2.0; mean difference 0.30, permutation $p = 0.003$). **The effect is small: Cohen's $d = 0.28$, and the median difference is not significant ($p = 0.053$).** We report it as preliminary. Its direction is nevertheless interpretable: a bypass is a missing signal, which steering supplies cheaply, whereas an override is a gating failure that additional signal does not repair.

The distinction bears on the design of steering-based defences. Uniform steering degrades benign performance, which has motivated adaptive schemes,[12] null-space constraints that preserve behaviour on benign inputs,[13] and activation scaling targeted at specific tokens.[14] Our results suggest a further axis of adaptation. The strength required depends on which mechanism failed, and that can be measured before the intervention is applied.

5 Negative and null results

Decoding choice does not affect success. Success rates under greedy and sampled decoding differ by at most 0.027 across all seven attacks. Reported success rates are commonly given under a single decoding configuration; on this model and these attacks, that choice is not a source of variance.

Refusal-head suppression does not distinguish the mechanisms. The refusal-writing heads are sharply localised at layer 10, heads 27, 5 and 12, in line with reports of attention-head specialisation for harmful features.[10] Suppression, however, does not track mechanism.

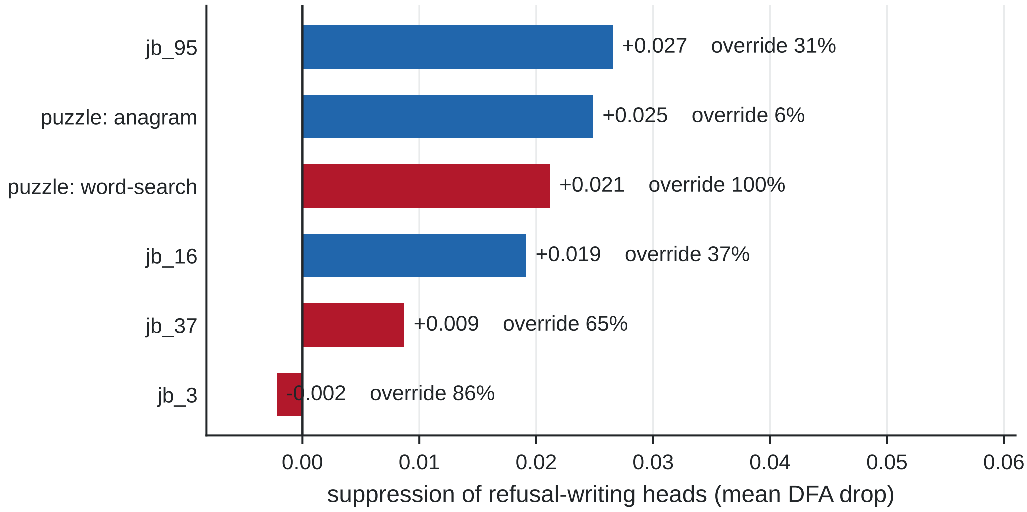

Figure 12: Suppression of refusal-writing heads by attack, annotated with each attack's override share.

The two puzzle variants suppress the refusal heads almost identically (+0.021 and +0.025) despite sitting at opposite mechanistic poles, Meanwhile the attack with the *least* suppression (−0.002) is override-dominant. This rules out the hypothesis that attacks succeed by disabling the refusal heads and locates the divergence downstream of them. It is consistent with the observation that harmful features remain robustly represented under attack.[10]

An activation-boundary defence was uninformative. Applying an activation-boundary defence to the classified successes restored refusal in 0% of cases in both classes, against a base rate of 1–2%. Nothing was rescued in either class, so no comparison between them is possible. We report this as a failed measurement, not as evidence of no effect.

A pre-registered prediction was falsified. Before running the steering experiment we predicted that override cases would be restored more cheaply than bypass cases, reasoning that refusal was already partly present to amplify. The opposite holds (Section 4.6). Our reinterpretation, that a missing signal is cheaper to supply than a broken gate is to repair, was formed after the fact and should be treated accordingly.

6 A prompt-rendering confound

We report this separately because it is a hazard for any pipeline that measures behaviour and activations in different components.

In an earlier configuration the two halves of the pipeline rendered prompts differently. One emitted a system block, empty when no system prompt was supplied; the other emitted none. The weights, the decoding strategy and the instruction text were identical; only the surrounding template differed.

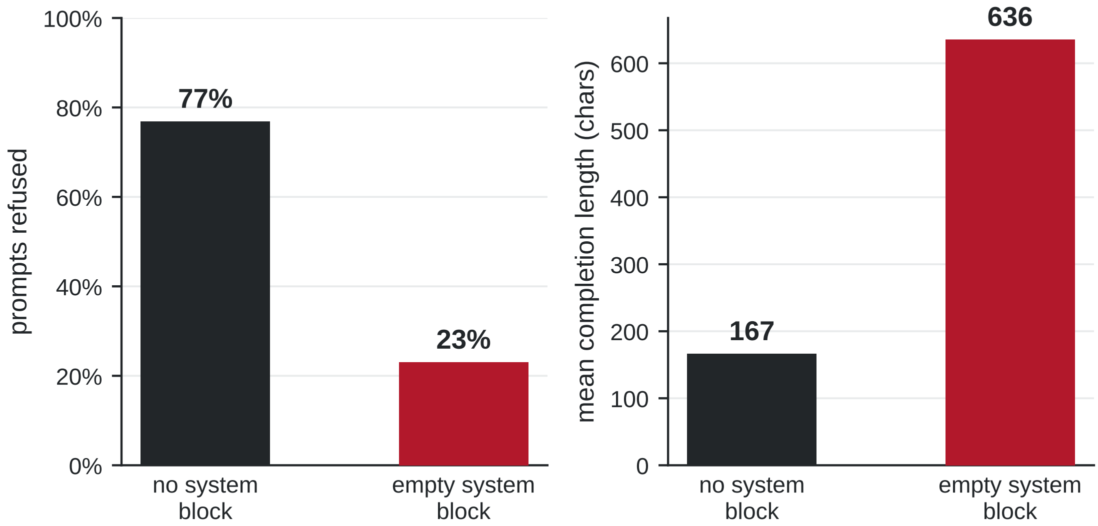

Figure 13: Refusal rate and mean completion length for the same 13 prompts under two renderings.

On 13 prompts previously judged successful, **7 flipped**, all in the same direction. Refusal fell from 10/13 to 3/13 and mean completion length rose from 167 to 636 characters. One prompt produced a 29-character refusal under one rendering and 1,391 characters of compliance under the other.

Activations had therefore been measured on the rendering the model refuses, while success labels came from the rendering it complies with. The two halves of the analysis described different prompts. The pipeline was rebuilt so that both components call a single rendering function, the variant is recorded in every output row, and a preflight check aborts when the two configurations disagree.

This warrants a general caution. Prompt template details are rarely reported, and a difference as small as an empty system block was sufficient here to invert the outcome on the majority of a test set.

7 Implementation

The measurement pipeline comprises approximately thirty modules distributed across two subsystems: one responsible for generation and judgement, and one for activation capture and mechanistic analysis. This section documents four design decisions that materially affected the validity or the cost of the results, and the testing regime applied throughout.

Verified template reconstruction. The persona attack templates were available only as pre-applied rewrites of ten behaviours, whereas the study required applying them to three hundred. The templates were therefore reconstructed by aligning rewrites of a common template across behaviours and extracting the invariant prefix and suffix. Reconstruction is accepted only if re-applying the recovered template reproduces every original rewrite exactly; any mismatch raises an exception. This enforces in code the constraint that no stimulus in the study is authored or approximated, which is otherwise a matter of discipline and therefore liable to erosion under time pressure.

Per-role provider configuration. The target and judge models are configured independently, permitting a single process to execute the target against local weights while dispatching judgement to a remote endpoint. This consolidated two previously separate subsystems into one execution path, which is the structural change that eliminated the prompt-rendering confound documented in Section 6.

Centralised prompt rendering. A single function renders every prompt in both subsystems. The rendering variant is recorded in each output row, and a preflight check terminates execution when the two subsystems are configured inconsistently. The module is among the smallest in the codebase and among the most consequential: the confound it prevents is invisible in the outputs and inverted the result on the majority of a test set.

Decoupling analysis from execution. Several components exist so that an expensive run can be reanalysed without repeating it. Classification thresholds are re-derived from stored projections, which do not depend on the threshold, and so require no GPU. Auxiliary probes are back-filled from stored completions. Rows on which the judge returned no parseable response are re-judged in isolation. Given that a full generation run occupies approximately thirty hours of GPU time, this separation avoided several complete re-executions over the course of the project.

Testing. Each substantive module carries a self-test executable on CPU against synthetic tensors, verifying arithmetic and control flow independently of model weights. Two defects were identified by this means before consuming GPU time: a scaling error that rendered a defence implementation degenerate, and a parameter combination that produced systematically invalid output. The regime does not verify numerical correctness against the model, but it establishes that the code paths execute as intended, which is the class of error most easily introduced during refactoring.

8 Limitations

A single model. All results are from Llama-3.1-8B-Instruct. Direction-based structure transfers across model families and scales,[5] but the specific mixture proportions reported here will likely vary.

Seven attacks from two families. The attacks are five persona templates and two puzzle variants. Optimisation-based attacks[15] and encoding-based attacks are not represented, and whether within-technique heterogeneity generalises to them is untested.

The bypass class is under-determined. "The refusal direction was not active" is consistent both with the model having recognised harm and not committed to refusing, and with the model not having recognised harm at all. HARC shows these are separable, near-orthogonal concepts.[5] We attempted a second, harmfulness direction and did not succeed.

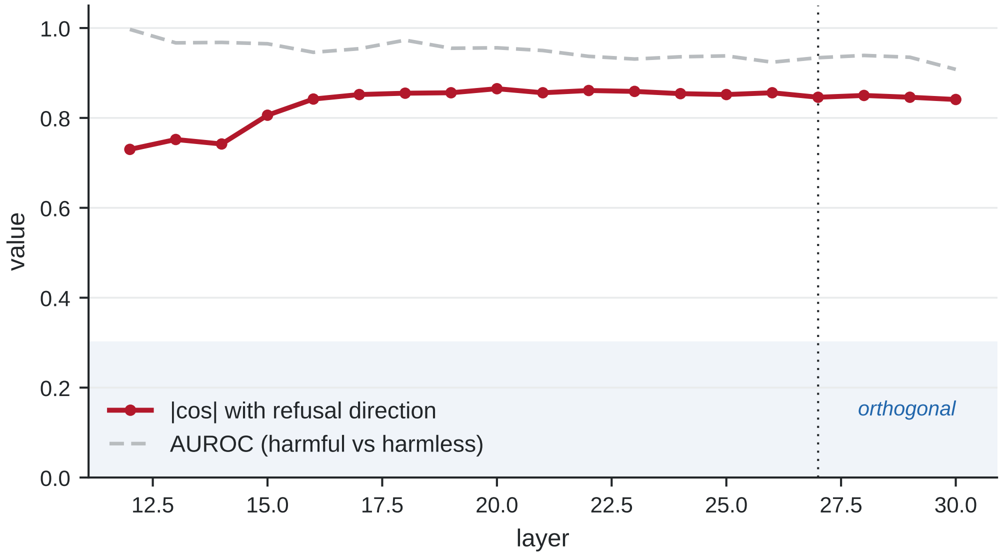

Figure 14: Absolute cosine between an extracted harmfulness direction and the refusal direction, by layer, with the harmful/harmless separation of the same direction shown for reference.

The extracted direction separates harmful from harmless prompts well (AUROC 0.91–1.00), but its cosine with the refusal direction remains between 0.73 and 0.86 at every layer from 12 to 30 and never approaches orthogonality. The cause is methodological: both directions were derived from the same harmful-versus-harmless contrast, differing only in token position, and a difference of means over a single contrast yields essentially one axis regardless of where it is read. Separable concepts require separable contrast sets. Consequently, every claim in this report concerns the activation of the refusal direction and not whether the model recognised harm.

Direction extracted from a subset. The refusal direction was extracted from a ten-behaviour pilot, not from the full 300, because the candidate sweep generates text for each of approximately 130 candidates and does not scale. The direction is causally validated, but it is calibrated on a small fraction of the data to which it is applied.

Attribution of mechanism to prompt properties is unresolved. Given that two puzzle variants diverge, a natural hypothesis is that some property of the prompt determines the mechanism. Obfuscation type does not predict it: the obfuscation category spans 6% to 100% override. A finer taxonomy based on whether harmful tokens appear literally, are recoverable by lookup, or require computation appears to predict perfectly, but each cell contains exactly one attack. With seven collected attacks, each varying many properties at once, attribution is not possible. Constructed stimuli varying one factor at a time would be required.

9 Conclusion

Attack success rate treats a jailbreak as a binary event. The internal record shows it is not: among 509 successful attacks on a single model, two distinct failures occur in roughly a two-to-one ratio, and the attack governs which occurs, not the request. Two variants of one technique, differing only in how they conceal the harmful words, sit at opposite extremes.

The practical consequence is that attack-level characterisation is a lossy summary of what a defence must handle. A defence tuned against an attack that overrides an active refusal representation is not thereby tuned against one that prevents the representation from activating.

Our steering results, though preliminary, indicate that the two respond differently to the same intervention.

The measurement apparatus itself proved a substantive source of error. A portion of a standard behaviour set elicits harmful content without any attack; a widely used judge fails silently on the most harmful outputs; and an unreported difference in prompt template inverted the outcome on the majority of a test set. These are properties of the evaluation stack, not of any model, and they apply to any work built on the same components.

References

[1] A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, N. Nanda. Refusal in language models is mediated by a single direction. *NeurIPS*, 2024. arXiv:2406.11717.

[2] M. Mazeika et al. HarmBench: a standardized evaluation framework for automated red teaming and robust refusal. *ICML*, 2024.

[3] A. Souly et al. A StrongREJECT for empty jailbreaks. *NeurIPS Datasets and Benchmarks*, 2024. arXiv:2402.10260.

[4] SoK: Robustness in large language models against jailbreak attacks. *IEEE Symposium on Security and Privacy*, 2026. arXiv:2605.05058.

[5] HARC: coupling harmfulness and refusal directions for robust safety alignment. arXiv:2607.00572, 2026.

[6] J. Ahn, K. Lee. PUZZLED: jailbreaking large language models through word-based puzzles. arXiv:2508.01306, 2025.

[7] Understanding jailbreak success: a study of latent space dynamics in large language models. arXiv:2406.09289, 2024.

[8] JailbreakLens: interpreting jailbreak mechanism in the lens of representation and circuit. arXiv:2411.11114, 2024.

[9] Jailbreak transferability emerges from shared representations. arXiv:2506.12913, 2025.

[10] Robust harmful features under jailbreak attacks: mechanistic evidence from attention head specialization in large language models. arXiv:2606.28153, 2026.

[11] Breaking refusal in the first half: a mechanistic study of the prefill jailbreak. arXiv:2607.14147, 2026.

[12] AdaSteer: your aligned LLM is inherently an adaptive jailbreak defender. arXiv:2504.09466, 2025.

[13] AlphaSteer: learning refusal steering with principled null-space constraint. arXiv:2506.07022, 2025.

[14] ASGuard: activation-scaling guard to mitigate targeted jailbreaking attack. arXiv:2509.25843, 2025.

[15] AdvPrefix: an objective for nuanced LLM jailbreaks. arXiv:2412.10321, 2024.